Fiscal Year:	FY 2007	**Task Last Updated:**	FY 07/24/2007
PI Name:	Turner, Russell T Ph.D.		
Project Title:	Pharmacological Intervention to Prevent Disuse Osteopenia- Oregon State		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Russell.Turner@oregonstate.edu	**Fax:**	FY 541 737 6914
PI Organization Type:	UNIVERSITY	**Phone:**	541 737 9545
Organization Name:	Oregon State University		
PI Address 1:	Nutrition & Exercise Sciences/Skeletal Biology Laboratory		
PI Address 2:	Milam Hall, Mail Stop 107B		
PI Web Page:			
City:	Corvallis	**State:**	OR
Zip Code:	97331-8558	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-01
Start Date:	04/25/2005	**End Date:**	10/31/2006
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 10/31/2006 per P. Denkins/JSC (3/07) PI moved from Mayo Clinic to Oregon. Received a new NAG #		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNJ05HD45G		
Performance Goal No.:			
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Task Description:	NOTE: Continuation of NAG9-1458 requiring new grant due to PI move from Mayo Clinic to Oregon State. We have previously obtained evidence suggesting that mechanical loading acts on bone cells via a cascade mechanism. We have postulated that the initial step of the cascade initiated by dynamic loading involves mechanoreceptor-transduced regulation of expression of a small number of immediate response genes which code for transcription factors and other cell cycle regulatory factors. These factors regulate expression of genes which code for skeletal signaling molecules (e.g., growth factors) which in turn influence the expression of genes to control cell proliferation, metabolism and expression of differentiated bone cell function. If this model is correct, disuse osteopenia is largely mediated by altered expression of cell signaling molecules and may be corrected by pharmacological agents which modulate the skeletal expression of these intermediates. Studies performed during the previous funding interval have revealed that the signal transduction pathways for parathyroid hormone (PTH) and estrogen partially overlap with weight bearing in that each of these agents regulates the expression of skeletal signaling molecules. These studies also demonstrated an unexpected pronounced gender difference in the skeletal response to decreased mechanical usage. The proposed research is designed to test the hypothesis that it is possible to bypass the mechanoreceptor step in hindlimb unloaded rats by pharmacological intervention with PTH and a tissue-selective estrogen agonist (clomiphene) with little activity on reproductive tissues. The specific aims of the proposed studies in hindlimb elevated rats are to determine: 1) the comparative effects of skeletal unloading on normal and gonadectomized males and females; 2) the efficacy of clomiphene treatment to prevent cancellous osteopenia and maintain normal bone turnover, architecture and mechanical properties; and 3) if the skeletal efficacy of combined treatment with clomiphene and human PTH (hPTH) is superior to monotherapy.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	All of our studies were related to Risk 1: “Accelerated bone loss and fracture risk”, in the Critical Path Roadmap. Question 1c: “Is there an additive or synergistic effect of gonadal hormone deficiency in men or women on bone loss during prolonged exposure to hypogravity?” Our studies have helped to answer this question by showing that: Estrogen deficiency and hypogravity have additive effects on bone loss in rats. Also, skeletal unloading resulted in bone loss in males and females by gender-independent mechanisms, increased bone resorption and decreased bone formation in both. However, our results also demonstrated that most of the bone loss in rats subjected to microgravity is due to decreased bone formation in animals with normal estrogen levels whereas it is due to increased bone resorption in estrogen deficient rats. As a consequence, the ideal drug (or exercise) intervention may have to take into consideration the gonadal hormone status individual astronauts. There is compelling evidence that estrogen sensitizes the skeleton to mechanical loading. We have investigated the possible contributions of metabolites of 17ß-estradiol, to the estrogen effect, especially those which are important following the menopause. These studies have revealed that 16 alpha-hydroxyestrone is a complete estrogen agonist on bone but a partial agonist on other estrogen target tissues. We have also shown that Tibolone, a synthetic steroid which undergoes tissue selective metabolism into several metabolites having estrogenic, progestogenic or androgenic activities, has direct beneficial effects on bone cells. We demonstrated gender differences in bone mass and bone turnover in adult rats. These differences (e.g., females have a higher rate of bone turnover than males) are similar to what we have observed in human iliac crest bone biopsies. Future studies will be needed to establish whether there are gender specific differences in the rates of bone loss. Additionally, studies should be performed to establish whether there are gender differences in the rates of recovery during hypogravity. Question 1d: “What pharmacological agent(s) will most effectively minimize the decrease in bone mass with extended exposure to hypogravity?” We have helped answer this question by performing dose response experiments which have shown that intermittent PTH can completely prevent the pronounced inhibitory effects of hindlimb microgravity on bone formation as well as prevent bone loss. However, PTH was only effective in preventing the detrimental skeletal effects of hindlimb unloading at dose rates that greatly exceed those used in humans. In contrast, therapeutic dose rates of PTH were highly bone anabolic in weight bearing limbs. Also, the requirement of PTH for weight bearing is much greater for cortical than cancellous bone. We are continuing to investigate whether combining PTH with SERMs has additional beneficial effects on the skeleton during hypogravity. We have investigated further the mechanisms mediating the detrimental side effects of PTH. This is important because side effects are likely to occur in some individuals receiving PTH therapy to prevent bone loss and the incidence of side effects might occur during hypogravity. We obtained evidence that the catabolic bone effects of PTH require PDGF-A signaling. Antagonizing this signaling with an inhibitor of PDGF-A expression and receptor binding blunted PTH-induced bone marrow fibrosis, hypercalcemia and bone resorption. Question 1h: “Does the hypogravity environment change the nutritional requirements for optimal bone health?” We have obtained data that strongly suggest that the hypogravity environment changes the response to “life style factors” and nutritional factors. We have investigated how selected “life style factors” such as alcohol consumption alter the skeletal response to mechanical unloading. These studies demonstrated that alcohol and hindlimb unloading have independent detrimental effects on bone formation. It is unlikely that astronauts will be consuming alcohol during spaceflight. However, our data suggests that detrimental skeletal effects of consuming alcohol persist when drinking ceases and could potentially exaggerate the effects of hypogravity. As mentioned above, we have compelling evidence that weight bearing and PTH act in synergy to increase bone formation. The level of PTH in blood is greatly influenced by ionic calcium. As a consequence, the circulating levels of PTH during hypogravity may be inadequate to maintain normal bone formation. Question 1j: “What systemic adaptation s to hypogravity are important contributing factors to bone loss, evaluations of which are essential to effective countermeasure development?” This question was not an original goal of the proposed research. However, we have now accumulated strong evidence that the mast cell functions as a critical accessory cell in mediating the detrimental skeletal effects of hyperparathyroidism, estrogen deficiency and inflammation. Our future

	efforts will include investigating the potential role of mast cells in mediating hypogravity-induced bone loss. As a starting point, we will investigate the relationship between mast cell numbers in bone in response to hypogravity using archival studies. Depending on the results, future studies will be performed in mast-cell deficient animals. These studies have the potential to reveal a novel target for the development of pharmacological agents to prevent disuse osteopenia. Question 1k: “Are hypogravity-induced changes in bone density, geometry, and architecture reversible upon encountering partial gravity exposure, or on return to full gravity?” We have performed additional studies to further develop our model systems and ability to accurately measure bone density, geometry and architecture. These include establishing the precise age of epiphyseal closure in rats using micro-CT and developing cryostatic micro-CT imaging. Using the latter method, we are able to obtain high resolution images for measuring in 3-D geometry and architecture while preserving the integrity of RNA within the tissue. We have established a method whereby we can obtain RNA following imaging in order to investigate gene expression in precise regions of bone. Our future efforts will focus on establishing the skeletal changes resulting from a return to full gravity following prolonged microgravity. Question 1l: “What regimen will most effectively hasten restoration of bone mass and bone strength to pre-flight values in returning crewmembers?” We have investigated whether vibration accelerates the restoration of bone mass and muscle function following hind limb unloading. We are in the process of analyzing the results.
Bibliography Type:	Description: (Last Updated: 01/12/2026)
Abstracts for Journals and Proceedings	Hefferan TE, Evans GL, Zhang M, Morey-Holton E, Turner RT. "The anabolic bone response to low dose PTH is attenuated by reduced physical activity. " 26th Annual Meeting of the American Society for Bone and Mineral Research, October 2004. J Bone Min Res 2004;19(Suppl) 1:S193. , Oct-2004
Abstracts for Journals and Proceedings	Hefferan TE, Evans GL, Morey-Holton E, Zhang M, Jewison D, Turner RT. "Hindlimb unloading attenuates the anabolic response to low dose parathyroid hormone. " 20th Annual Meeting of the American Society for Gravitational and Space Biology, New York, NY, November 2004. Gravitational and Space Biology Bulletin 2004 Nov;18(1):57. , Nov-2004
Abstracts for Journals and Proceedings	Maran A, Shogren KL, Zhang M, Yaszemski MJ, Turner RT. "Evidence that interferon signaling is required for 2-ME-mediated osteosarcoma cell death. " 27th Annual Meeting of the American Society for Bone and Mineral Research, October 2005. J Bone Miner Res 2005;20:S211. , Oct-2005
Articles in Peer-reviewed Journals	Turner RT, Evans GL, Lotinun S, Lapke PD, Iwaniec UT, Morey-Holton E. "Dose-response effects of intermittent PTH on cancellous bone in hindlimb unloaded rats." J Bone Miner Res. 2007 Jan;22(1):64-71. PMID: 17042715 , Jan-2007
Articles in Peer-reviewed Journals	Shogren KL, Turner RT, Yaszemski MJ, Maran A. "Double-stranded RNA-dependent protein kinase is involved in 2-methoxyestradiol-mediated cell death of osteosarcoma cells." J Bone Miner Res. 2007 Jan;22(1):29-36. PMID: 17014383 , Jan-2007
Articles in Peer-reviewed Journals	Maran A, Gorny G, Oursler MJ, Zhang M, Shogren KL, Yaszemski MJ, Turner RT. "2-methoxyestradiol inhibits differentiation and is cytotoxic to osteoclasts." J Cell Biochem. 2006 Oct 1;99(2):425-34. PMID: 16619269 , Oct-2006
Articles in Peer-reviewed Journals	Bliziotes M, Sibonga JD, Turner RT, Orwoll E. "Periosteal remodeling at the femoral neck in nonhuman primates." J Bone Miner Res. 2006 Jul;21(7):1060-7. PMID: 16813526 , Jul-2006
Articles in Peer-reviewed Journals	Turner RT, Lotinun S, Hefferan TE, Morey-Holton E. "Disuse in adult male rats attenuates the bone anabolic response to a therapeutic dose of parathyroid hormone." J Appl Physiol. 2006 Sep;101(3):881-6. PMID: 16675609 , Sep-2006
Articles in Peer-reviewed Journals	Lotinun S, Sibonga JD, Turner RT. "Evidence that the cells responsible for marrow fibrosis in a rat model for hyperparathyroidism are preosteoblasts." Endocrinology. 2005 Sep;146(9):4074-81. PMID: 15947001 , Sep-2005
Articles in Peer-reviewed Journals	Onyia JE, Helvering LM, Gelbert L, Wei T, Huang S, Chen P, Dow ER, Maran A, Zhang M, Lotinun S, Lin X, Halladay DL, Miles RR, Kulkarni NH, Ambrose EM, Ma YL, Frolik CA, Sato M, Bryant HU, Turner RT. "Molecular profile of catabolic versus anabolic treatment regimens of parathyroid hormone (PTH) in rat bone: an analysis by DNA microarray." J Cell Biochem. 2005 May 15;95(2):403-18. PMID: 15779007 , May-2005
Articles in Peer-reviewed Journals	Smith BJ, Lucas EA, Turner RT, Evans GL, Lerner MR, Brackett DJ, Stoecker BJ, Arjmandi BH. "Vitamin E provides protection for bone in mature hindlimb unloaded male rats." Calcif Tissue Int. 2005 Apr;76(4):272-9. PMID: 15742232 , Apr-2005
Articles in Peer-reviewed Journals	Turner RT, Iwaniec UT. "Low dose parathyroid hormone maintains normal bone formation in adult male rats during rapid weight loss." Bone. 2011 Apr 1;48(4):726-32. http://dx.doi.org/10.1016/j.bone.2010.12.034 ; PubMed PMID: 21215827 , Apr-2011
Articles in Peer-reviewed Journals	Iwaniec UT, Turner RT, Smith BJ, Stoecker BJ, Rust A, Zhang B, Vasu VT, Gohil K, Cross CE, Traber MG. "Evaluation of long-term vitamin E insufficiency or excess on bone mass, density, and microarchitecture in rodents." Free Radic Biol Med. 2013 Dec;65:1209-14. http://dx.doi.org/10.1016/j.freeradbiomed.2013.09.004 ; PubMed PMID: 24051180 , Dec-2013